Image Classification and Transfer Learning

Multiclass outputs, augmentation, and pretrained features

ECE 116 | Fall 2026

Adapted from Farias and Ramakrishnan, MIT OpenCourseWare 15.773, Spring 2024. Changes were made. CC BY-NC-SA 4.0.

Representing Images Digitally

How Grayscale Images are Represented



	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]	[,11]	[,12]	[,13]	[,14]	[,15]	[,16]	[,17]	[,18]	[,19]	[,20]	[,21]	[,22]	[,23]	[,24]	[,25]	[,26]	[,27]	[,28]
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[2,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[3,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[4,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[5,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[6,]	0	0	0	0	0	0	0	0	0	0	0	0	3	18	18	18	126	136	175	26	166	255	247	127	0	0	0	0
[7,]	0	0	0	0	0	0	0	0	30	36	94	154	170	253	253	253	253	253	225	172	253	242	195	64	0	0	0	0
[8,]	0	0	0	0	0	0	0	49	238	253	253	253	253	253	253	253	253	251	93	82	82	56	39	0	0	0	0	0
[9,]	0	0	0	0	0	0	0	18	219	253	253	253	253	253	198	182	247	241	0	0	0	0	0	0	0	0	0	0
[10,]	0	0	0	0	0	0	0	0	80	156	107	253	253	205	11	0	43	154	0	0	0	0	0	0	0	0	0	0
[11,]	0	0	0	0	0	0	0	0	0	14	1	154	253	90	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[12,]	0	0	0	0	0	0	0	0	0	0	0	139	253	190	2	0	0	0	0	0	0	0	0	0	0	0	0	0
[13,]	0	0	0	0	0	0	0	0	0	0	0	11	190	253	70	0	0	0	0	0	0	0	0	0	0	0	0	0
[14,]	0	0	0	0	0	0	0	0	0	0	0	0	35	241	225	160	108	1	0	0	0	0	0	0	0	0	0	0
[15,]	0	0	0	0	0	0	0	0	0	0	0	0	0	81	240	253	253	119	25	0	0	0	0	0	0	0	0	0
[16,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	45	186	253	253	150	27	0	0	0	0	0	0	0	0
[17,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	16	93	252	253	187	0	0	0	0	0	0	0	0
[18,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	249	253	249	64	0	0	0	0	0	0	0
[19,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	46	130	183	253	253	207	2	0	0	0	0	0	0	0
[20,]	0	0	0	0	0	0	0	0	0	0	0	0	39	148	229	253	253	253	250	182	0	0	0	0	0	0	0	0
[21,]	0	0	0	0	0	0	0	0	0	0	24	114	221	253	253	253	253	201	78	0	0	0	0	0	0	0	0	0
[22,]	0	0	0	0	0	0	0	0	23	66	213	253	253	253	253	198	81	2	0	0	0	0	0	0	0	0	0	0
[23,]	0	0	0	0	0	0	18	171	219	253	253	253	253	195	80	9	0	0	0	0	0	0	0	0	0	0	0	0
[24,]	0	0	0	0	55	172	226	253	253	253	253	244	133	11	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[25,]	0	0	0	0	136	253	253	253	212	135	132	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[26,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[27,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[28,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

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How Grayscale Images are Represented



	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]	[,11]	[,12]	[,13]	[,14]	[,15]	[,16]	[,17]	[,18]	[,19]	[,20]	[,21]	[,22]	[,23]	[,24]	[,25]	[,26]	[,27]	[,28]
[1,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[2,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[3,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[4,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[5,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[6,]	0	0	0	0	0	0	0	0	0	0	0	0	3	18	18	18	126	136	175	26	166	255	247	127	0	0	0	0
[7,]	0	0	0	0	0	0	0	0	30	36	94	154	170	253	253	253	253	253	225	172	253	242	195	64	0	0	0	0
[8,]	0	0	0	0	0	0	0	49	238	253	253	253	253	253	253	253	253	251	93	82	82	56	39	0	0	0	0	0
[9,]	0	0	0	0	0	0	0	18	219	253	253	253	253	253	198	182	247	241	0	0	0	0	0	0	0	0	0	0
[10,]	0	0	0	0	0	0	0	0	80	156	107	253	253	205	11	0	43	154	0	0	0	0	0	0	0	0	0	0
[11,]	0	0	0	0	0	0	0	0	0	14	1	154	253	90	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[12,]	0	0	0	0	0	0	0	0	0	0	0	139	253	190	2	0	0	0	0	0	0	0	0	0	0	0	0	0
[13,]	0	0	0	0	0	0	0	0	0	0	0	11	190	253	70	0	0	0	0	0	0	0	0	0	0	0	0	0
[14,]	0	0	0	0	0	0	0	0	0	0	0	0	35	241	225	160	108	1	0	0	0	0	0	0	0	0	0	0
[15,]	0	0	0	0	0	0	0	0	0	0	0	0	0	81	240	253	253	119	25	0	0	0	0	0	0	0	0	0
[16,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	45	186	253	253	150	27	0	0	0	0	0	0	0	0
[17,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	16	93	252	253	187	0	0	0	0	0	0	0	0
[18,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	249	253	249	64	0	0	0	0	0	0	0
[19,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	46	130	183	253	253	207	2	0	0	0	0	0	0	0
[20,]	0	0	0	0	0	0	0	0	0	0	0	0	39	148	229	253	253	253	250	182	0	0	0	0	0	0	0	0
[21,]	0	0	0	0	0	0	0	0	0	0	24	114	221	253	253	253	253	201	78	0	0	0	0	0	0	0	0	0
[22,]	0	0	0	0	0	0	0	0	23	66	213	253	253	253	253	198	81	2	0	0	0	0	0	0	0	0	0	0
[23,]	0	0	0	0	0	0	18	171	219	253	253	253	253	195	80	9	0	0	0	0	0	0	0	0	0	0	0	0
[24,]	0	0	0	0	55	172	226	253	253	253	253	244	133	11	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[25,]	0	0	0	0	136	253	253	253	212	135	132	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[26,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[27,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
[28,]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

- A grayscale image is a rectangular array of pixels
- The light intensity of each pixel is a number between 0 and 255. As the number increases from 0 to 255, the pixel goes from black through gray to white
- Each cell of the matrix shows the light intensity of the pixel at that location

Grayscale light intensity matrix © fast.ai. All rights reserved. This content is excluded from our Creative Commons license. For more information, see https://ocw.mit.edu/help/faq-fair-use.

How Color Images are Represented

- Each pixel of a color image is represented by three intensities (not one), corresponding to the pixel's "redness", "blueness" and "greenness" (RGB)

- Each light intensity is still a number between 0 and 255

- Thus color images are represented as 3 matrices of numbers, corresponding to the Red, Green and Blue "channels" respectively.

How Color Images are Represented



killian.jpeg
JPEG image - 6 KB

Tags	Add Tags...
Created	Today, 4:30 PM
Modified	Today, 4:30 PM
Content created	Friday, April 3, 2020 at 4:30 PM
Dimensions	200×200
Color space	RGB

Red

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]
[1,]	147	131	138	144	131	134	144	135	133	145
[2,]	140	131	141	149	138	138	143	132	136	146
[3,]										
[4,]										
[5,]										
[6,]										
[7,]										
[8,]										
[9,]										
[10,]										

Green

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]
[1,]	186	171	179	185	171	172	180	171	168	180
[2,]	177	169	180	188	176	175	178	167	169	180
[3,]										
[4,]										
[5,]										
[6,]										
[7,]										
[8,]										
[9,]										
[10,]										

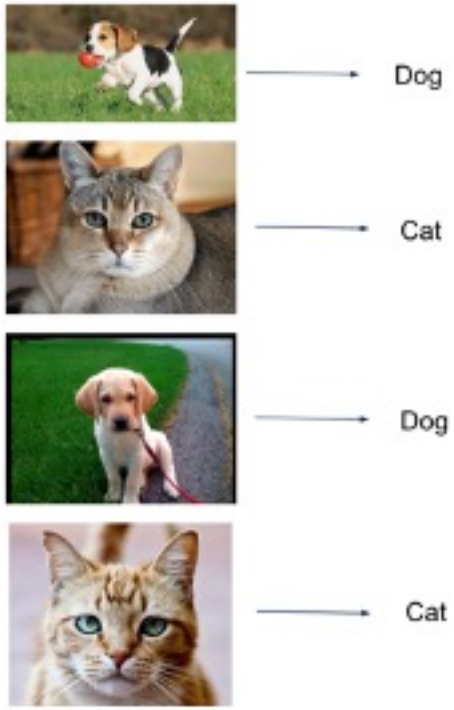
Blue

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]
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[2,]	248	234	239	245	238	246	255	251	246	243
[3,]	255	241	238	236	229	241	253	249	238	234
[4,]	255	252	243	233	228	237	242	234	218	205
[5,]	255	255	249	231	228	231	224	215	204	166
[6,]	255	255	230	192	189	202	205	205	204	147
[7,]	231	231	188	140	138	152	156	159	177	136
[8,]	155	172	149	114	113	111	93	82	119	115
[9,]	107	130	108	93	113	100	67	66	81	95
[10,]	84	104	90	69	69	61	52	63	59	46

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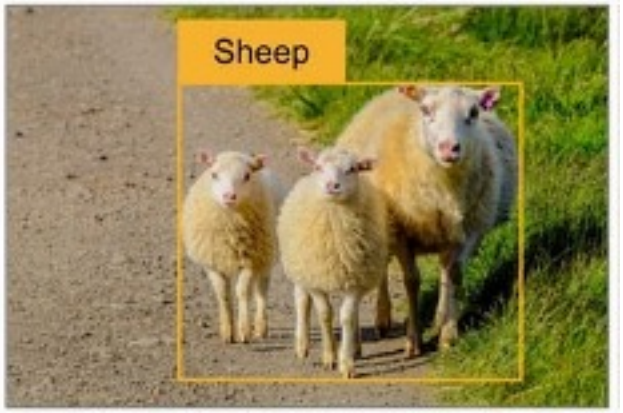
Key Tasks in Computer Vision

Image Classification



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Classification and Localization

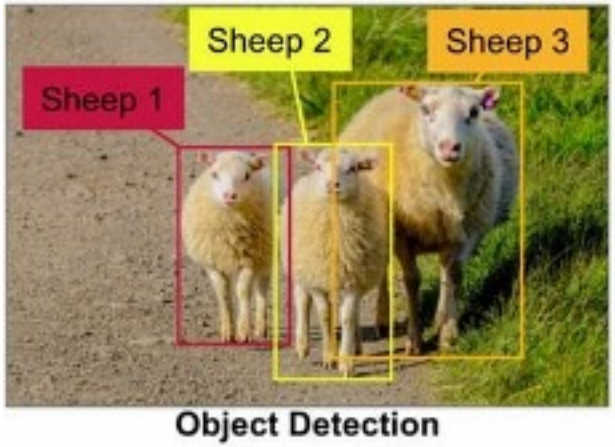


Classification + Localization

Sheep image © Nirmala Murali on Medium. All rights reserved. This content is excluded from our Creative Commons license. For more information, see https://ocw.mit.edu/help/faq-fair-use.

Image Source

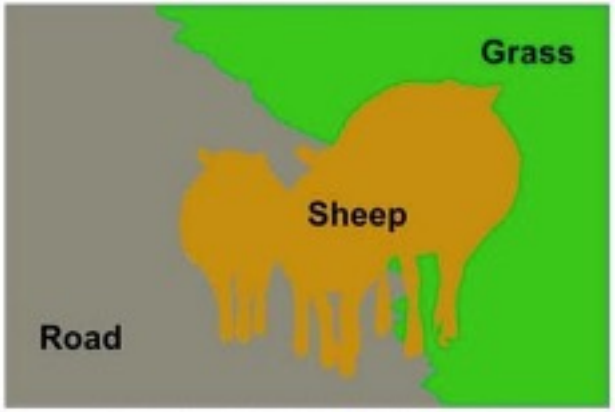
Object Detection



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Image Source

Semantic Segmentation

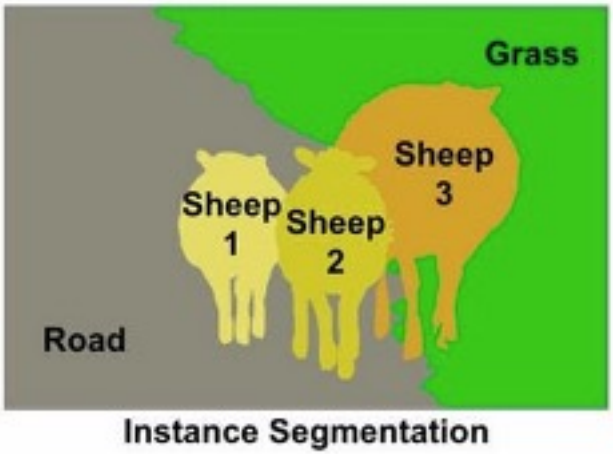


Sheep image © Nirmala Murali on Medium. All rights reserved. This content is excluded from our Creative Commons license. For more information, see https://ocw.mit.edu/help/faq-fair-use.

Every pixel needs to be classified into one of N categories

Image Source

Instance Segmentation



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Every pixel needs to be classified into one of N categories and

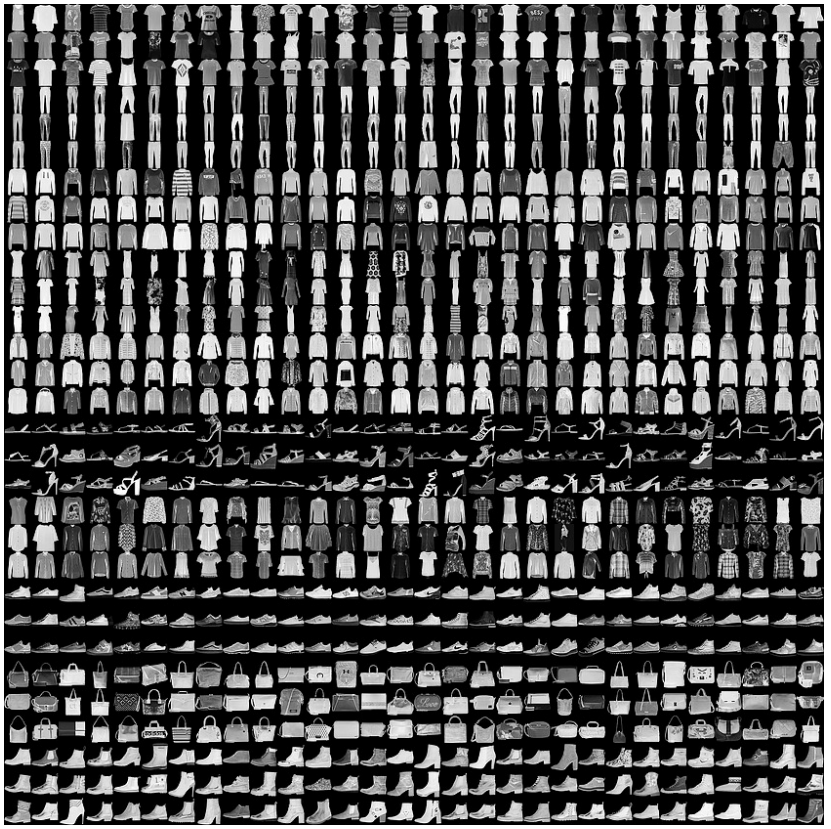
Different instances (e.g., Sheep 1, Sheep 2, Sheep 3) of the same category (e.g., Sheep) need to be identified

Image Source

Image Classification

Motivating application: Fashion MNIST

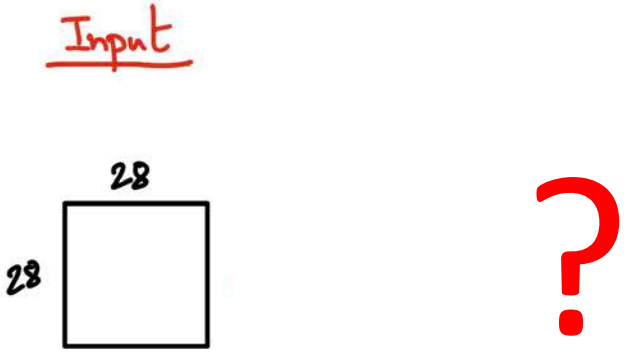
The fashion-mnist dataset consists of 70,000 images of clothing items across 10 categories.



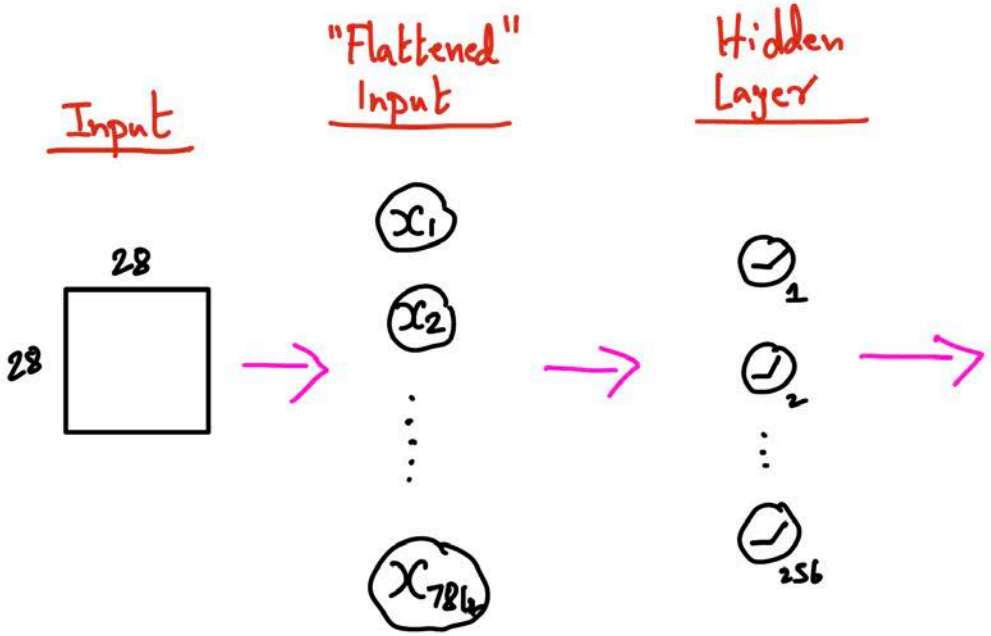
Sample of the Fashion MNIST images by Yuzamei.
Source: Wikimedia Commons. License: CC BY-SA.

Source: https://www.kaggle.com/datasets/zalando-research/fashionmnist

A simple NN to classify grayscale clothing images



A simple NN to classify grayscale clothing images

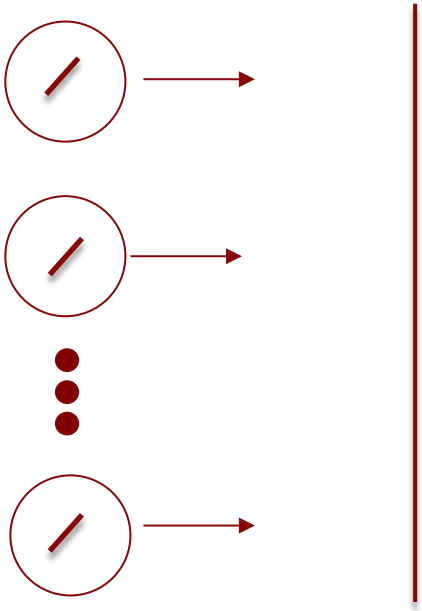


How do we represent a 10-way output?

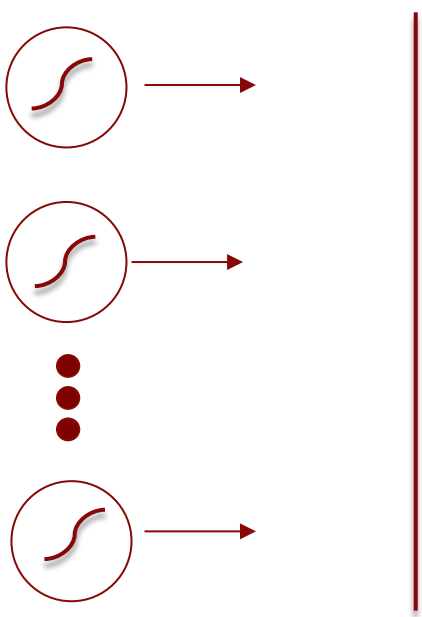
Multi-Class Classification

Suppose the output variable is categorical with 10 levels

We know how to output 10 numbers



We know how to output 10 probabilities



How do we output 10 probabilities that sum to 1.0?

The Softmax Layer

softmax takes in *n* arbitrary numbers and converts them to *n* probabilities



Summary: Output Layers for Regression and Classification

Output Variable	Output Layer
Single number (regression with a single output)	
Single probability (binary classification)	
Vector of *n* numbers (regression with multiple outputs)	Stack of 
Vector of n probabilities that add up to 1 (multi-class classification)	Softmax

Class-label encoding

Encoding is how the dataset stores the correct class for an image.

Fashion MNIST example: Sneaker

Integer label (sparse)	One-hot label
7	**[0,0,0,0,0,0,0,1,0,0]**

Both labels identify the same class for the same image.

The number 7 is a class index, not a measure of how sneaker-like it is.

Loss function and label format

The network produces the same 10 softmax probabilities either way.



10 output classes; class 7 is Sneaker

Correct label stored as 7

Correct label stored as a vector

`[0,0,0,0,0,0,0,1,0,0]`

Keras loss:

`sparse_categorical_crossentropy`

Keras loss:

`categorical_crossentropy`

Both losses use the predicted probability for class 7.

Binary case: one sigmoid output and a 0/1 label use binary_crossentropy.

Training a Model with Limited Data

Web-scraped dataset of < 100 color images of handbags and shoes

With this tiny dataset, how do we will build a deep learning network to classify your shoe or handbag in class with high accuracy!



Generic images of handbags and shoes © unknown. All rights reserved. This content is excluded from our Creative Commons license. For more information, see https://ocw.mit.edu/help/faq-fair-use.

Can we do better? We only have 100 examples of each class

Training augmentation preserves the label



Original

handbag



Horizontal flip

handbag



Mild brightness change

handbag

Use plausible changes that keep the class label valid.

Apply augmentation during training. Keep validation and test images fixed.

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Transfer Learning with Pre-trained Networks

Transfer learning takes advantage of two research trends

Trend 1: Researchers have designed NN architectures that are well-matched to different *types* of data. For example:

Type of data	Architecture
All	Residual connections
Images	Convolutional layers
Sequences (e.g., natural language, audio, video, gene sequences)	Transformers

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Trend 2: Using these architectural innovations, researchers have trained high-performance DNNs on a variety of large real-world datasets. Numerous pretrained models are available!

Transfer learning involves **customizing** such a pre-trained network to your problem, rather than designing and training a network from scratch.

Can we apply Transfer Learning to build a better Handbags/Shoes Classifier?*

Handbags and shoes are “everyday objects” and you can look around and see if there are any networks that have been trained on a dataset of images of “everyday objects”

*The first thing to do, of course, is to see if someone has already built such a classifier and put it on GitHub ☺

Can we apply Transfer Learning to build a better Handbags/Shoes Classifier?

Handbags and shoes are “everyday objects” and you can look around and see if there are any networks that have been trained on a dataset of images of “everyday objects”

Turns out the ImageNet dataset has millions of images of 1000 categories of everyday objects. We can look for networks that have been trained on ImageNet

IMAGENET

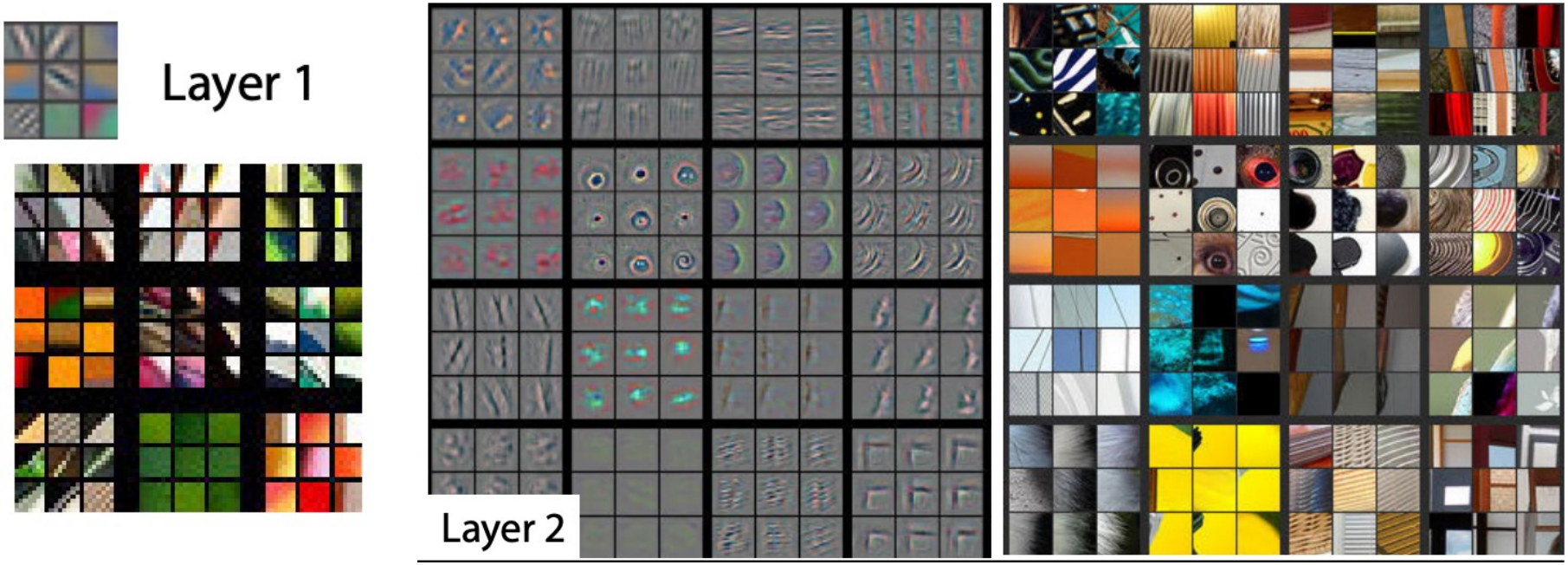
- 1,000 object classes (categories).
- Images:
  - 1.2 M train
  - 100k test.



https://www.slideshare.net/xavigiro/image-classification-on-imagenet-d1l4-2017-upc-deep-learning-for-computer-vision/

ImageNet training image © unknown. All rights reserved. This content is excluded from our Creative Commons license. For more information, see https://ocw.mit.edu/help/faq-fair-use.

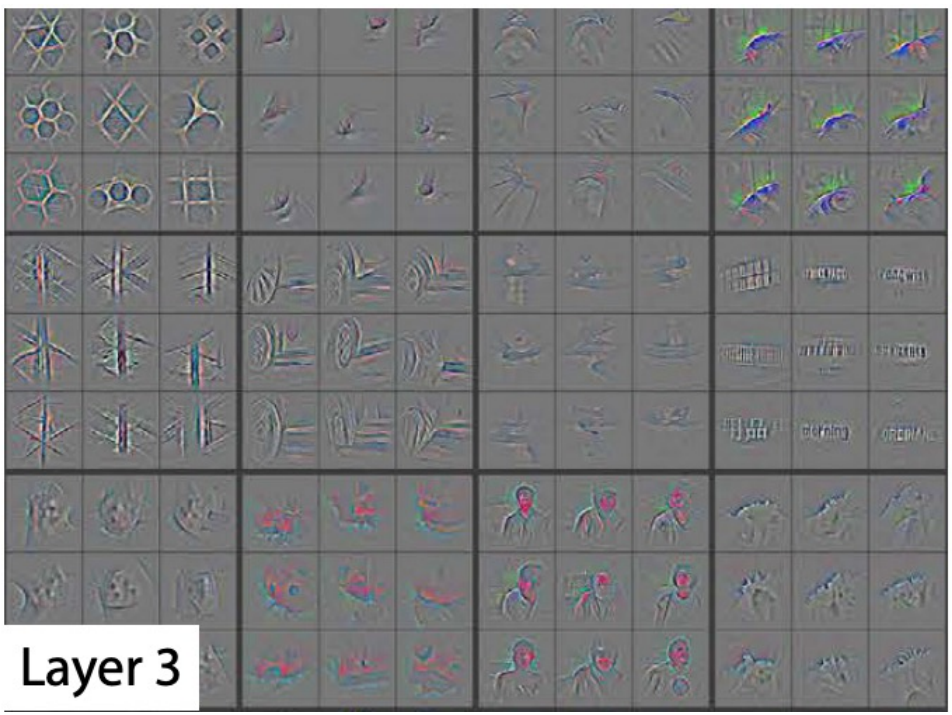
A network trained on ImageNet and that does well on it has essentially developed a smart, hierarchical representation of these objects across its layers



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https://cs.nyu.edu/~fergus/papers/zeilerECCV2014.pdf

A network trained on ImageNet and that does well on it has essentially developed a smart, hierarchical representation of these objects across its layers

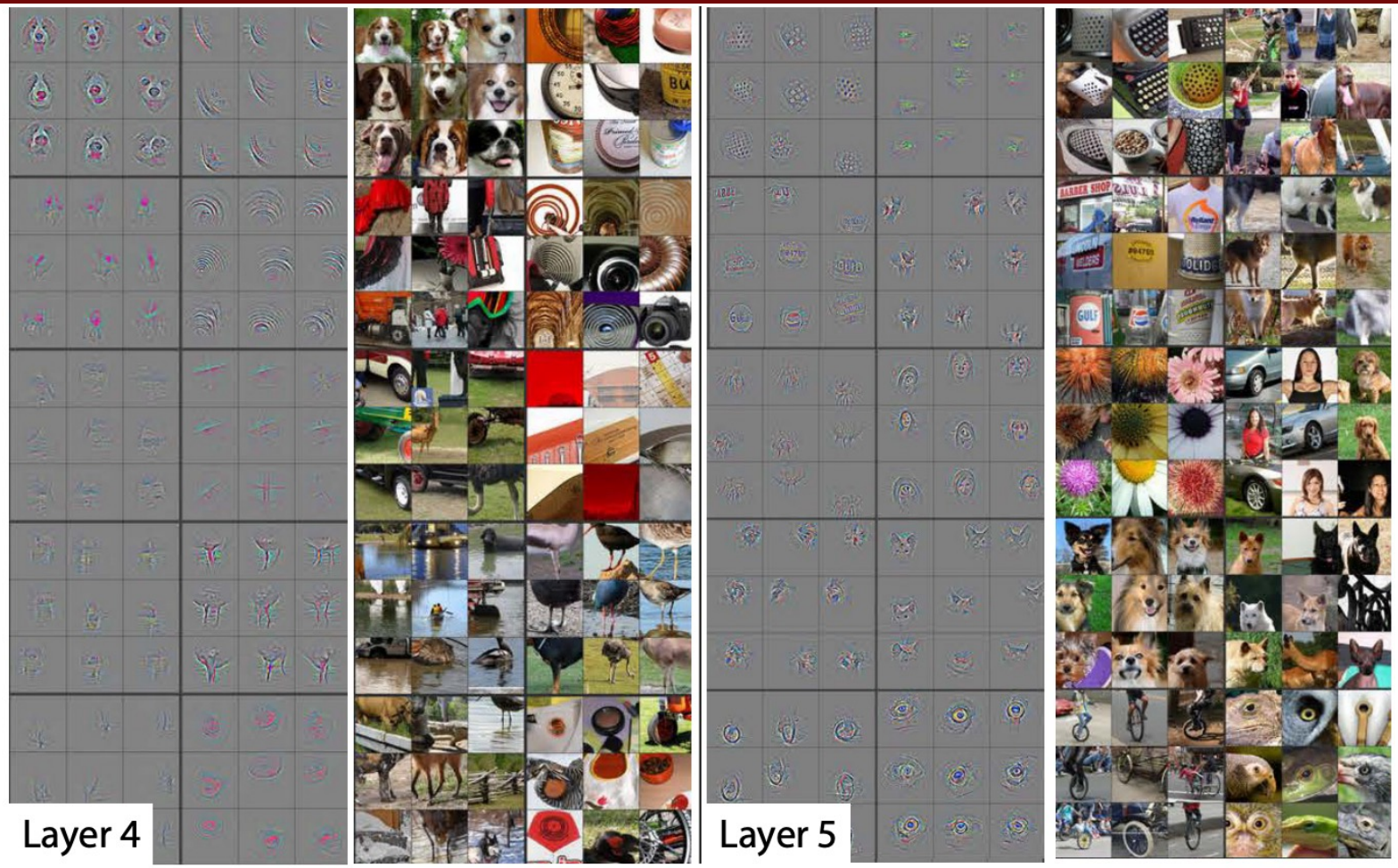




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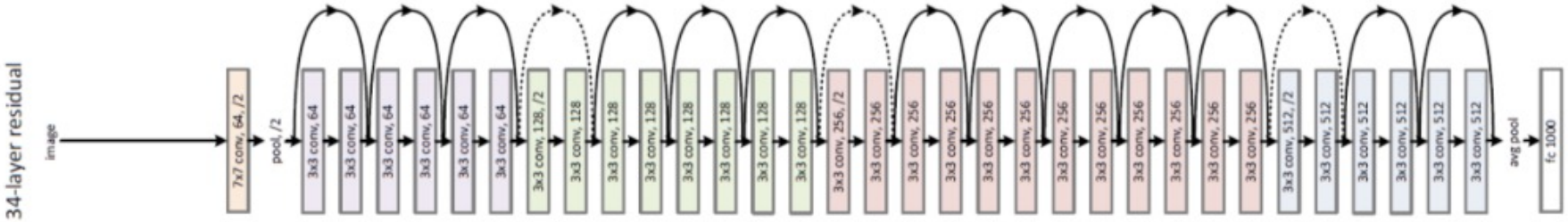


https://cs.nyu.edu/~fergus/papers/zeilerECCV2014.pdf

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The ResNet family of networks was trained on ImageNet and did very well in the associated ImageNet competition

ResNet 34*



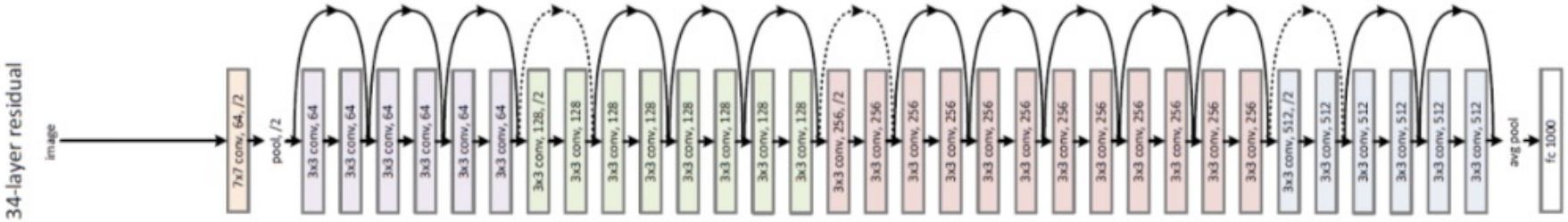
We'd expect the (learned) weights and biases of ResNet layers to embody "knowledge" (like illustrated in the previous slides) about the characteristics of the millions of everyday images that the network was trained on

ResNet34 layers image © Kaiming He, Xiangyu Zhang, Shaoqing Ren, Jian Sun/arXiv. All rights reserved. This content is excluded from our Creative Commons license. For more information, see https://ocw.mit.edu/help/faq-fair-use.

*https://arxiv.org/abs/1512.03385

But we can't use ResNet as is

ResNet 34*



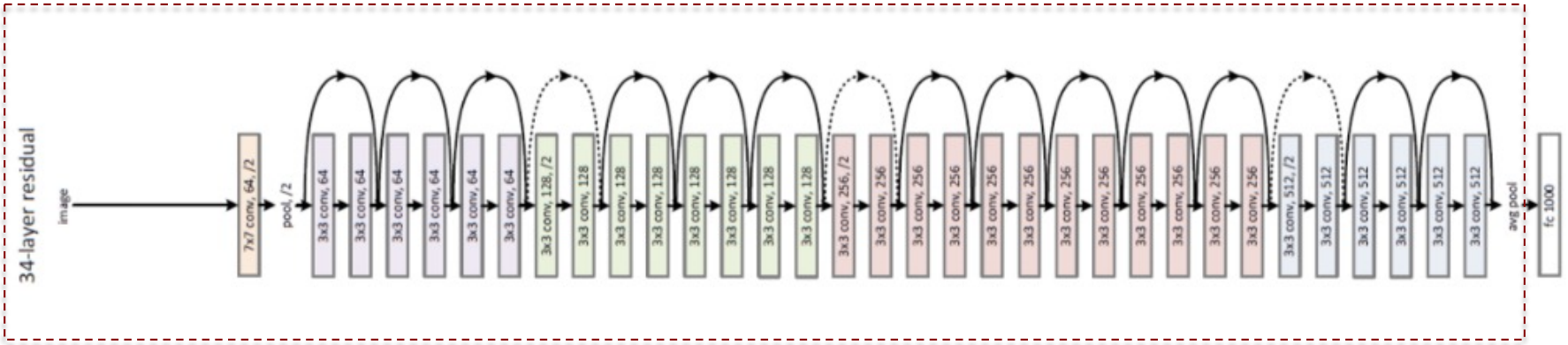
Remember that ResNet was designed to classify the image into 1000 categories. Our problem is different – we only care about two categories: handbags and shoes.

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So we take ResNet and stop just before the last layer

ResNet 34

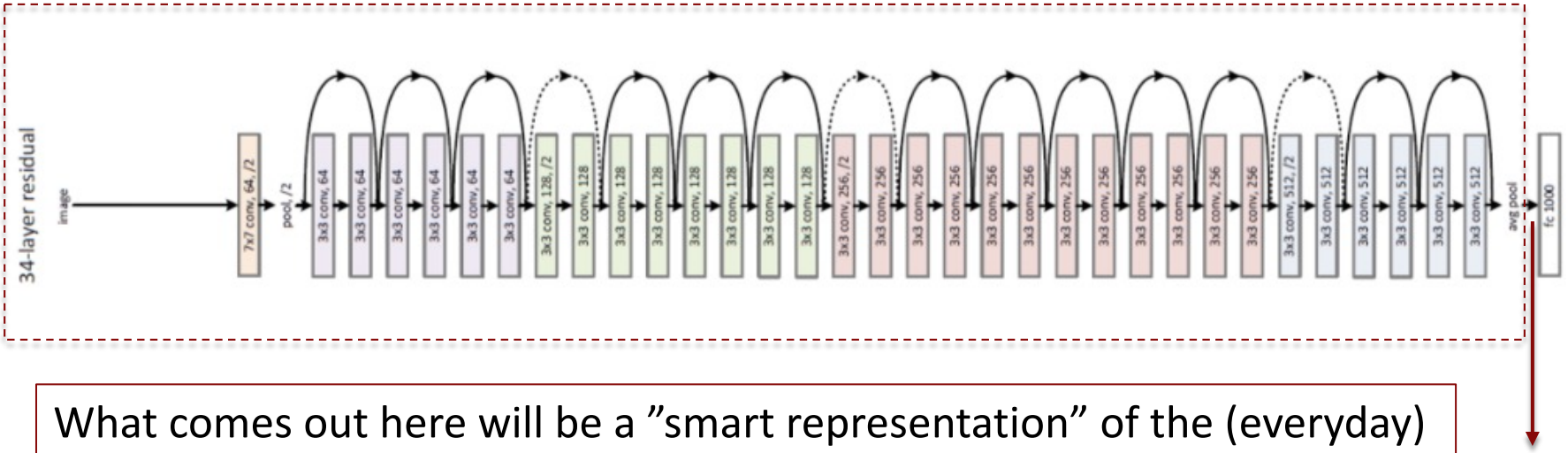


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*https://arxiv.org/abs/1512.03385

We can run our images through this "headless" ResNet

ResNet 34



What comes out here will be a "smart representation" of the (everyday) image that can be used for different kinds of categorization.

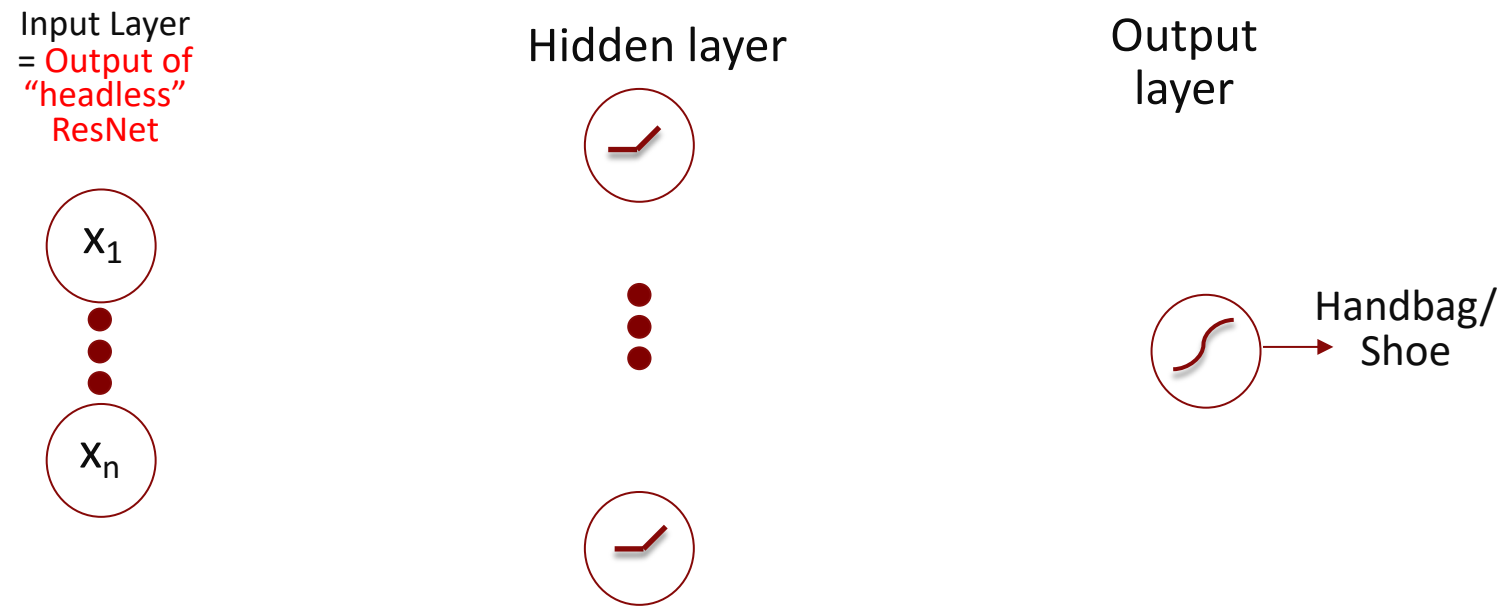
Imagine that the image gets "tagged" with lots of general tags and we can use the tags any way we wish.

*https://arxiv.org/abs/1512.03385

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We can think of the output of “headless ResNet” as a smart representation of the raw image and use it to train a simple one-hidden-layer NN

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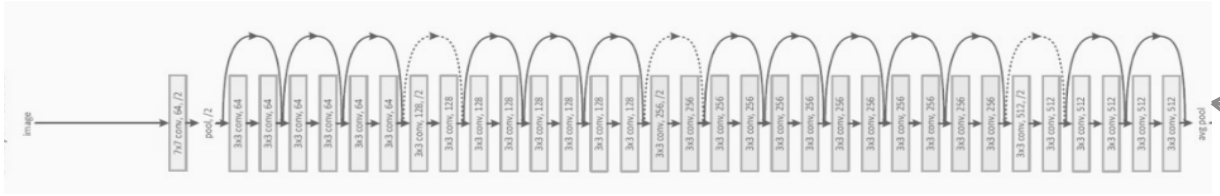
Since the inputs to the hidden layer are at a higher level of abstraction, it may be able to learn to classify handbags from shoes with very few examples

That’s the basic idea behind transfer learning …

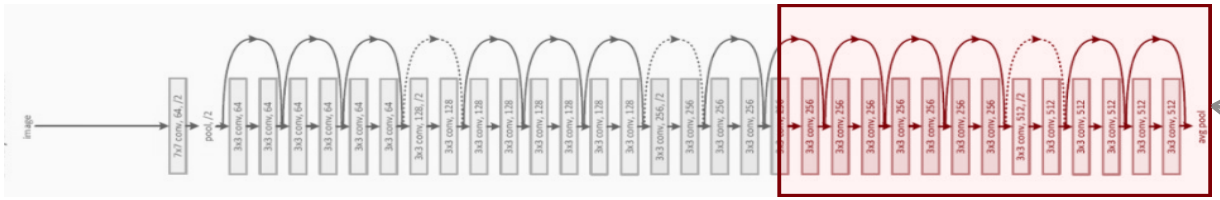
Trainable layers in a pretrained CNN

pretrained backbone | new head

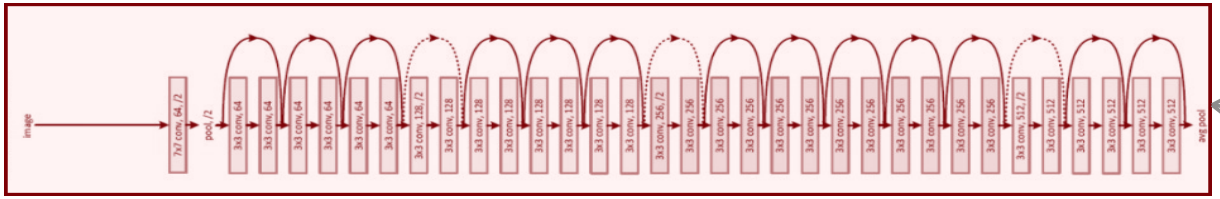
Frozen backbone



Partial fine-tuning



Full fine-tuning



Each approach starts from pretrained weights. The new head always trains.

trainable layers frozen layers

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This lecture adapts material from:

Vivek Farias and Rama Ramakrishnan, Hands-On Deep Learning,
MIT OpenCourseWare 15.773, Spring 2024.

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Changes include course framing, sequencing, omissions, and ECE116 branding.